

## Electronic Circuits

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## FETs vs. BJTs

### Similarities:

- Amplifiers
- Switching devices
- Impedance matching circuits

### Differences:

- FETs are voltage controlled devices.
- BJTs are current controlled devices.
- FETs have a higher input impedance.
- BJTs have higher gains.
- FETs are less sensitive to temperature variations and are more easily integrated on ICs.
- FETs are generally more static sensitive than BJTs.

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## FET Types

• **JFET:** Junction FET

• **MOSFET:** Metal-Oxide-Semiconductor FET

• **D-MOSFET:** Depletion MOSFET

• **E-MOSFET:** Enhancement MOSFET

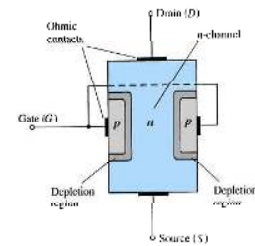
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## JFET Construction

There are two types of JFETs

- **n-channel**
- **p-channel**

The n-channel is more widely used.



There are three terminals:

- **Drain (D)** and **Source (S)** are connected to the n-channel
- **Gate (G)** is connected to the p-type material

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## JFET Operation: The Basic Idea

JFET operation can be compared to a water spigot.

**The source** of water pressure is the accumulation of electrons at the negative pole of the drain-source voltage.

**The drain** of water is the electron deficiency (or holes) at the positive pole of the applied voltage.

**The control** of flow of water is the gate voltage that controls the width of the n-channel and, therefore, the flow of charges from source to drain.



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## JFET Operating Characteristics

There are three basic operating conditions for a JFET:

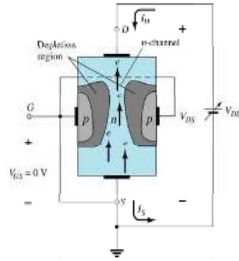
- $V_{GS} = 0$ ,  $V_{DS}$  increasing to some positive value
- $V_{GS} < 0$ ,  $V_{DS}$  at some positive value
- Voltage-controlled resistor

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## JFET Operating Characteristics: $V_{GS} = 0\text{ V}$

Three things happen when  $V_{GS} = 0$  and  $V_{DS}$  is increased from 0 to a more positive voltage

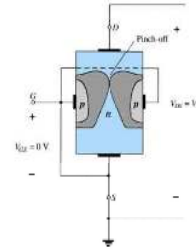
- The depletion region between p-gate and n-channel increases as electrons from n-channel combine with holes from p-gate.
- Increasing the depletion region, decreases the size of the n-channel which increases the resistance of the n-channel.
- Even though the n-channel resistance is increasing, the current ( $I_D$ ) from source to drain through the n-channel is increasing. This is because  $V_{DS}$  is increasing.



## JFET Operating Characteristics: Pinch Off

If  $V_{GS} = 0$  and  $V_{DS}$  is further increased to a more positive voltage, then the depletion zone gets so large that it **pinches off** the n-channel.

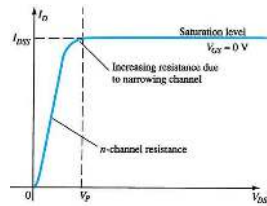
This suggests that the current in the n-channel ( $I_D$ ) would drop to 0A, but it does just the opposite—as  $V_{DS}$  increases, so does  $I_D$ .



## JFET Operating Characteristics: Saturation

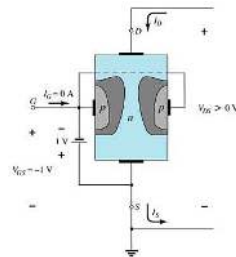
At the pinch-off point:

- Any further increase in  $V_{GS}$  does not produce any increase in  $I_D$ .  $V_{GS}$  at pinch-off is denoted as  $V_P$ .
- $I_D$  is at saturation or maximum. It is referred to as  $I_{DSS}$ .
- The ohmic value of the channel is maximum.



## JFET Operating Characteristics

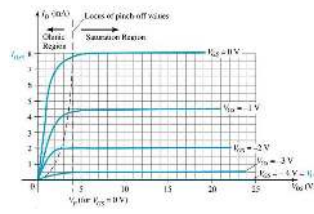
As  $V_{GS}$  becomes more negative, the depletion region increases.



## JFET Operating Characteristics

As  $V_{GS}$  becomes more negative:

- The JFET experiences pinch-off at a lower voltage ( $V_P$ ).
- $I_D$  decreases ( $I_D < I_{DSS}$ ) even though  $V_{DS}$  is increased.
- Eventually  $I_D$  reaches 0 A.  $V_{GS}$  at this point is called  $V_P$  or  $V_{GS(off)}$ .



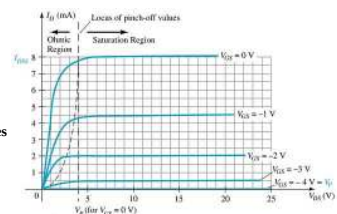
Also note that at high levels of  $V_{DS}$  the JFET reaches a breakdown situation.  $I_D$  increases uncontrollably if  $V_{DS} > V_{DSmax}$ .

## JFET Operating Characteristics: Voltage-Controlled Resistor

The region to the left of the pinch-off point is called the **ohmic region**.

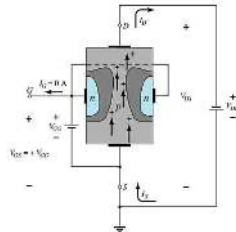
The JFET can be used as a variable resistor, where  $V_{GS}$  controls the drain-source resistance ( $r_d$ ). As  $V_{GS}$  becomes more negative, the resistance ( $r_d$ ) increases.

$$r_d = \frac{r_0}{\left(1 - \frac{V_{GS}}{V_P}\right)^2}$$



## p-Channel JFETS

The p-channel JFET behaves the same as the n-channel JFET, except the voltage polarities and current directions are reversed.

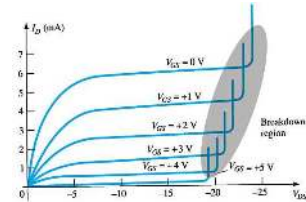


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## p-Channel JFET Characteristics

As  $V_{GS}$  increases more positively

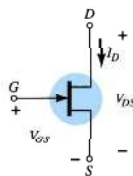
- The depletion zone increases
- $I_D$  decreases ( $I_D < I_{DSS}$ )
- Eventually  $I_D = 0$  A



Also note that at high levels of  $V_{DS}$  the JFET reaches a breakdown situation:  $I_D$  increases uncontrollably if  $V_{DS} > V_{DSmax}$ .

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## N-Channel JFET Symbol



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## JFET Transfer Characteristics

The transfer characteristic of input-to-output is not as straightforward in a JFET as it is in a BJT.

In a BJT,  $\beta$  indicates the relationship between  $I_B$  (input) and  $I_C$  (output).

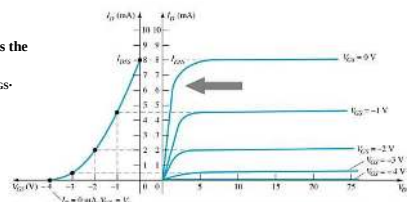
In a JFET, the relationship of  $V_{GS}$  (input) and  $I_D$  (output) is a little more complicated:

$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$$

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## JFET Transfer Curve

This graph shows the value of  $I_D$  for a given value of  $V_{GS}$ .



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## Plotting the JFET Transfer Curve

Using  $I_{DSS}$  and  $V_p$  ( $V_{GS(off)}$ ) values found in a specification sheet, the transfer curve can be plotted according to these three steps:

**Step 1**

$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$$

Solving for  $V_{GS} = 0$   $I_D = I_{DSS}$

**Step 2**

$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$$

Solving for  $V_{GS} = V_p$  ( $V_{GS(off)}$ )  $I_D = 0$  A

**Step 3**

$$\text{Solving for } V_{GS} = 0 \text{ V to } V_p \quad I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_p} \right)^2$$

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## JFET Specifications Sheet

### Electrical Characteristics

| ELECTRICAL CHARACTERISTICS ( $T_A = 25^\circ\text{C}$ unless otherwise noted)                                                        |               |      |      |      |                  |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------|------|------|------|------------------|
| Characteristic                                                                                                                       | Symbol        | Min  | Typ  | Max  | Unit             |
| <b>OFF CHARACTERISTICS</b>                                                                                                           |               |      |      |      |                  |
| Gate-Source Breakdown Voltage<br>( $I_G = 0, V_{GS} = V_{DS}$ )                                                                      | $V_{GS(off)}$ | -25  | —    | —    | Vdc              |
| Gate-Source Current<br>( $V_{GS} = -15\text{ Vdc}, V_{DS} = 0$<br>( $V_{GS} = -15\text{ Vdc}, V_{DS} = 0, T_A = 100^\circ\text{C}$ ) | $I_{GSS}$     | —    | —    | -1.0 | $\mu\text{Adc}$  |
| Gate-Source Cutoff Voltage<br>( $V_{GS} = 15\text{ Vdc}, I_G = 15\text{ nAdc}$ )                                                     | $V_{GS(off)}$ | -0.5 | —    | -0.5 | Vdc              |
| Gate-Source Voltage<br>( $V_{GS} = 15\text{ Vdc}, I_G = 10\text{ } \mu\text{Adc}$ )                                                  | $V_{GS}$      | —    | -2.5 | —    | Vdc              |
| <b>ON CHARACTERISTICS</b>                                                                                                            |               |      |      |      |                  |
| On-State Voltage Drop Coefficient<br>( $V_{GS} = 15\text{ Vdc}, V_{DS} = 0$ )                                                        | $V_{DS(on)}$  | 1.0  | 1.0  | 2.0  | mVdc             |
| <b>SMALL-SIGNAL CHARACTERISTICS</b>                                                                                                  |               |      |      |      |                  |
| Forward Transfer Admittance Coefficient<br>( $V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1\text{ kHz}$ )                                | $Y_{fs}$      | —    | 1000 | —    | $\mu\text{mhos}$ |
| Output Admittance Coefficient<br>( $V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1\text{ kHz}$ )                                          | $Y_{os}$      | —    | 10   | 20   | $\mu\text{mhos}$ |
| Input Capacitance<br>( $V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1.0\text{ MHz}$ )                                                    | $C_{iss}$     | —    | 4.5  | 1.0  | pF               |
| Reverse Transfer Capacitance<br>( $V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1.0\text{ MHz}$ )                                         | $C_{rss}$     | —    | 1.5  | 2.0  | pF               |

(Values are typical for 2N5457 and 2N5458)

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## JFET Specifications Sheet

### Maximum Ratings

| Rating                                                                                 | Symbol    | Value       | Unit             |
|----------------------------------------------------------------------------------------|-----------|-------------|------------------|
| Drain-Source Voltage                                                                   | $V_{DS}$  | 25          | Vdc              |
| Drain-Gate Voltage                                                                     | $V_{DG}$  | 25          | Vdc              |
| Reverse Gate-Source Voltage                                                            | $V_{GS}$  | -25         | Vdc              |
| Gate Current                                                                           | $I_G$     | 10          | mAdc             |
| Total Device Dissipation @ $T_A = 25^\circ\text{C}$<br>Derate above $25^\circ\text{C}$ | $P_D$     | 310         | mW               |
| Junction Temperature Range                                                             | $T_J$     | 125         | $^\circ\text{C}$ |
| Storage Channel Temperature Range                                                      | $T_{stg}$ | -55 to +150 | $^\circ\text{C}$ |

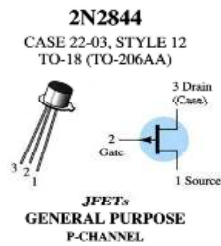


Refer to 2N4220 for graphs.

more...

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## Case and Terminal Identification



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## Testing JFETs

- **Curve Tracer**  
A curve tracer displays the  $I_D$  versus  $V_{DS}$  graph for various levels of  $V_{GS}$ .
- **Specialized FET Testers**  
These testers show  $I_{DSS}$  for the JFET under test.

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## MOSFETs

MOSFETs have characteristics similar to JFETs and additional characteristics that make them very useful.

There are two types of MOSFETs:

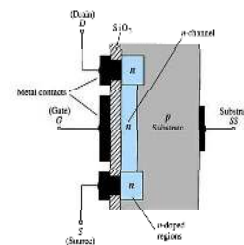
- Depletion-Type
- Enhancement-Type

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## Depletion-Type MOSFET Construction

The **Drain (D)** and **Source (S)** connect to the  $n$ -doped regions. These  $n$ -doped regions are connected via an  $n$ -channel. This  $n$ -channel is connected to the **Gate (G)** via a thin insulating layer of  $\text{SiO}_2$ .

The  $n$ -doped material lies on a  $p$ -doped substrate that may have an additional terminal connection called **Substrate (SS)**.

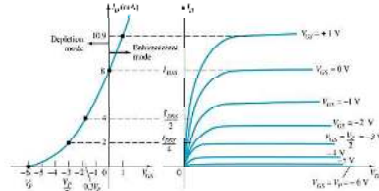


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## Basic MOSFET Operation

A depletion-type MOSFET can operate in two modes:

- Depletion mode
- Enhancement mode

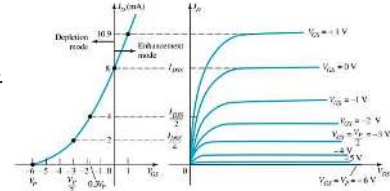


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## D-Type MOSFET in Depletion Mode

### Depletion Mode

The characteristics are similar to a JFET.



- When  $V_{GS} = 0$  V,  $I_D = I_{DSS}$
- When  $V_{GS} < 0$  V,  $I_D < I_{DSS}$
- The formula used to plot the transfer curve still applies:

$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_P} \right)^2$$

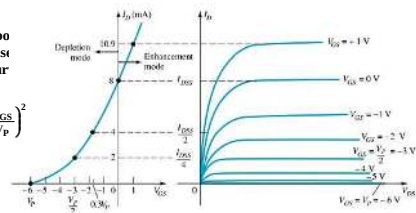
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## D-Type MOSFET in Enhancement Mode

### Enhancement Mode

- $V_{GS} > 0$  V
- $I_D$  increases above
- The formula used to plot the transfer curve still applies:

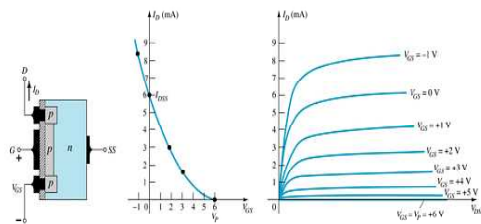
$$I_D = I_{DSS} \left( 1 - \frac{V_{GS}}{V_P} \right)^2$$



Note that  $V_{GS}$  is now a positive polarity

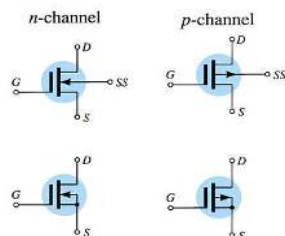
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## p-Channel D-Type MOSFET



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## D-Type MOSFET Symbols

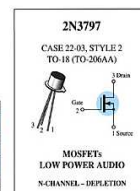


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## Specification Sheet

### Maximum Ratings

| Rating                                                                                | Symbol    | Value       | Unit             |
|---------------------------------------------------------------------------------------|-----------|-------------|------------------|
| Drain-Source Voltage                                                                  | $V_{DS}$  | 20          | Vdc              |
| Gate-Source Voltage                                                                   | $V_{GS}$  | 110         | Vdc              |
| Drain Current                                                                         | $I_D$     | 20          | mA               |
| Total Device Dissipation @ $T_A = 25^\circ\text{C}$ , Derate above $25^\circ\text{C}$ | $P_D$     | 200         | mW               |
| Storage Temperature Range                                                             | $T_{STG}$ | -55 to +200 | $^\circ\text{C}$ |



more...

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## Specification Sheet

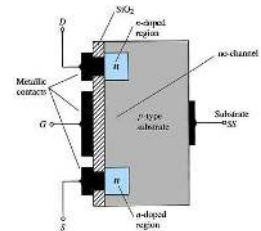
### Electrical Characteristics

| ELECTRICAL CHARACTERISTICS (T <sub>amb</sub> = 25°C unless otherwise indicated)                                                                         |         |                     |      |      |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------|------|------|------|
| Characteristics                                                                                                                                         |         |                     |      |      |      |
| <b>OFF CHARACTERISTICS</b>                                                                                                                              |         |                     |      |      |      |
| Static Drain-Source Leakage Voltage<br>(V <sub>GS</sub> = 0V, V <sub>DS</sub> = 50V)                                                                    | 250/300 | V <sub>DS(ON)</sub> | 20   | 25   | —    |
| Gate-Source Capacitance (C <sub>iss</sub> )<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 0V)<br>(f = 1MHz, V <sub>GS</sub> = 0V, V <sub>DS</sub> = 10V) | —       | C <sub>iss</sub>    | —    | 18   | 2000 |
| Drain-Source Capacitance (C <sub>oss</sub> )<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V)                                                          | 250/300 | V <sub>DS(ON)</sub> | —    | —    | 18   |
| Drain-Source Capacitance (C <sub>oss</sub> )<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V)                                                          | —       | C <sub>oss</sub>    | —    | —    | 18   |
| <b>ON CHARACTERISTICS</b>                                                                                                                               |         |                     |      |      |      |
| Static Drain-Source Current<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V)                                                                           | 250/300 | I <sub>DS</sub>     | 1.0  | 2.0  | 0.0  |
| Static Drain-Source Current<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V)                                                                           | 250/300 | I <sub>DS</sub>     | 0.0  | 1.0  | 10   |
| <b>SMALL SIGNAL CHARACTERISTICS</b>                                                                                                                     |         |                     |      |      |      |
| Transfer Function<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V, f = 1 kHz)                                                                          | 250/300 | g <sub>m</sub>      | 1000 | 1000 | 1000 |
| Output Admittance<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V, f = 1 kHz)                                                                          | 250/300 | h <sub>oe</sub>     | —    | —    | —    |
| Input Capacitance<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V, f = 1 kHz)                                                                          | 250/300 | C <sub>iss</sub>    | —    | 18   | 2000 |
| Reverse Transfer Capacitance<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V, f = 1 kHz)                                                               | 250/300 | C <sub>oss</sub>    | —    | 18   | 2000 |
| <b>ESD PROTECTION CHARACTERISTICS</b>                                                                                                                   |         |                     |      |      |      |
| Static Drain-Source Current<br>(V <sub>GS</sub> = 10V, V <sub>DS</sub> = 50V, f = 1 kHz, V <sub>DS</sub> = 50V)                                         | 250/300 | I <sub>DS</sub>     | —    | —    | —    |

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## E-Type MOSFET Construction

- The **Drain (D)** and **Source (S)** connect to the *n*-doped regions. These *n*-doped regions are connected via an *n*-channel
- The **Gate (G)** connects to the *p*-doped substrate via a thin insulating layer of SiO<sub>2</sub>
- There is no channel
- The *n*-doped material lies on a *p*-doped substrate that may have an additional terminal connection called the **Substrate (SS)**

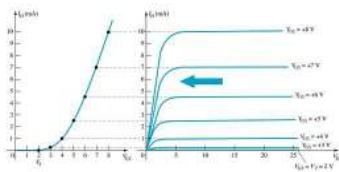


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## Basic Operation of the E-Type MOSFET

The enhancement-type MOSFET operates only in the enhancement mode.

- V<sub>GS</sub> is always positive
- As V<sub>GS</sub> increases, I<sub>D</sub> increases
- As V<sub>GS</sub> is kept constant and V<sub>DS</sub> is increased, then I<sub>D</sub> saturates (I<sub>DSS</sub>) and the saturation level, V<sub>DSsat</sub> is reached



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## E-Type MOSFET Transfer Curve

To determine I<sub>D</sub> given V<sub>GS</sub>:

$$I_D = k(V_{GS} - V_T)^2$$

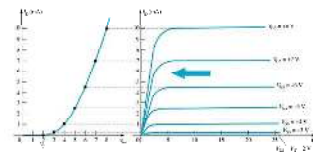
Where:  
V<sub>T</sub> = threshold voltage or voltage at which the MOSFET turns on

k, a constant, can be determined by using values at a specific point and the formula:

$$k = \frac{I_{D(ON)}}{(V_{GS(ON)} - V_T)^2}$$

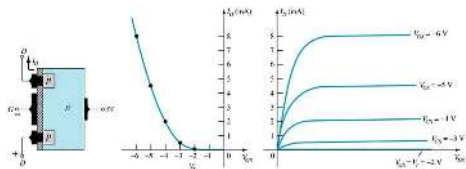
V<sub>DSsat</sub> can be calculated by:

$$V_{DSat} = V_{GS} - V_T$$



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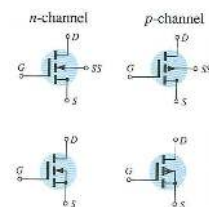
## p-Channel E-Type MOSFETs



The *p*-channel enhancement-type MOSFET is similar to the *n*-channel, except that the voltage polarities and current directions are reversed.

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## MOSFET Symbols



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# Specification Sheet

## Maximum Ratings

| Symbol                                                 | Unit | Value                | Unit |
|--------------------------------------------------------|------|----------------------|------|
| Rated power (continuous)                               | W    | 100                  | Watt |
| Rated voltage                                          | V    | 250                  | Volt |
| Rated current (average)                                | A    | 0.4                  | Amp  |
| Rated current (peak)                                   | A    | 0.8                  | Amp  |
| Rated current (RMS)                                    | A    | 0.4                  | Amp  |
| Rated current (RMS) (100% duty cycle)                  | A    | 0.4                  | Amp  |
| Rated current (RMS) (50% duty cycle)                   | A    | 0.8                  | Amp  |
| Rated current (RMS) (25% duty cycle)                   | A    | 1.6                  | Amp  |
| Rated current (RMS) (10% duty cycle)                   | A    | 3.2                  | Amp  |
| Rated current (RMS) (5% duty cycle)                    | A    | 6.4                  | Amp  |
| Rated current (RMS) (2.5% duty cycle)                  | A    | 12.8                 | Amp  |
| Rated current (RMS) (1% duty cycle)                    | A    | 25.6                 | Amp  |
| Rated current (RMS) (0.5% duty cycle)                  | A    | 51.2                 | Amp  |
| Rated current (RMS) (0.25% duty cycle)                 | A    | 102.4                | Amp  |
| Rated current (RMS) (0.1% duty cycle)                  | A    | 204.8                | Amp  |
| Rated current (RMS) (0.05% duty cycle)                 | A    | 409.6                | Amp  |
| Rated current (RMS) (0.025% duty cycle)                | A    | 819.2                | Amp  |
| Rated current (RMS) (0.01% duty cycle)                 | A    | 1638.4               | Amp  |
| Rated current (RMS) (0.005% duty cycle)                | A    | 3276.8               | Amp  |
| Rated current (RMS) (0.0025% duty cycle)               | A    | 6553.6               | Amp  |
| Rated current (RMS) (0.001% duty cycle)                | A    | 13107.2              | Amp  |
| Rated current (RMS) (0.0005% duty cycle)               | A    | 26214.4              | Amp  |
| Rated current (RMS) (0.00025% duty cycle)              | A    | 52428.8              | Amp  |
| Rated current (RMS) (0.0001% duty cycle)               | A    | 104857.6             | Amp  |
| Rated current (RMS) (0.00005% duty cycle)              | A    | 209715.2             | Amp  |
| Rated current (RMS) (0.000025% duty cycle)             | A    | 419430.4             | Amp  |
| Rated current (RMS) (0.00001% duty cycle)              | A    | 838860.8             | Amp  |
| Rated current (RMS) (0.000005% duty cycle)             | A    | 1677721.6            | Amp  |
| Rated current (RMS) (0.0000025% duty cycle)            | A    | 3355443.2            | Amp  |
| Rated current (RMS) (0.000001% duty cycle)             | A    | 6710886.4            | Amp  |
| Rated current (RMS) (0.0000005% duty cycle)            | A    | 13421772.8           | Amp  |
| Rated current (RMS) (0.00000025% duty cycle)           | A    | 26843545.6           | Amp  |
| Rated current (RMS) (0.0000001% duty cycle)            | A    | 53687091.2           | Amp  |
| Rated current (RMS) (0.00000005% duty cycle)           | A    | 107374182.4          | Amp  |
| Rated current (RMS) (0.000000025% duty cycle)          | A    | 214748364.8          | Amp  |
| Rated current (RMS) (0.00000001% duty cycle)           | A    | 429496729.6          | Amp  |
| Rated current (RMS) (0.000000005% duty cycle)          | A    | 858993459.2          | Amp  |
| Rated current (RMS) (0.0000000025% duty cycle)         | A    | 1717986918.4         | Amp  |
| Rated current (RMS) (0.000000001% duty cycle)          | A    | 3435973836.8         | Amp  |
| Rated current (RMS) (0.0000000005% duty cycle)         | A    | 6871947673.6         | Amp  |
| Rated current (RMS) (0.00000000025% duty cycle)        | A    | 13743895347.2        | Amp  |
| Rated current (RMS) (0.0000000001% duty cycle)         | A    | 27487790694.4        | Amp  |
| Rated current (RMS) (0.00000000005% duty cycle)        | A    | 54975581388.8        | Amp  |
| Rated current (RMS) (0.000000000025% duty cycle)       | A    | 109951162777.6       | Amp  |
| Rated current (RMS) (0.00000000001% duty cycle)        | A    | 219902325555.2       | Amp  |
| Rated current (RMS) (0.000000000005% duty cycle)       | A    | 439804651110.4       | Amp  |
| Rated current (RMS) (0.0000000000025% duty cycle)      | A    | 879609302220.8       | Amp  |
| Rated current (RMS) (0.000000000001% duty cycle)       | A    | 1759218604441.6      | Amp  |
| Rated current (RMS) (0.0000000000005% duty cycle)      | A    | 3518437208883.2      | Amp  |
| Rated current (RMS) (0.00000000000025% duty cycle)     | A    | 7036874417766.4      | Amp  |
| Rated current (RMS) (0.0000000000001% duty cycle)      | A    | 14073748835532.8     | Amp  |
| Rated current (RMS) (0.00000000000005% duty cycle)     | A    | 28147497671065.6     | Amp  |
| Rated current (RMS) (0.000000000000025% duty cycle)    | A    | 56294995342131.2     | Amp  |
| Rated current (RMS) (0.00000000000001% duty cycle)     | A    | 112589990684262.4    | Amp  |
| Rated current (RMS) (0.000000000000005% duty cycle)    | A    | 225179981368524.8    | Amp  |
| Rated current (RMS) (0.0000000000000025% duty cycle)   | A    | 450359962737049.6    | Amp  |
| Rated current (RMS) (0.000000000000001% duty cycle)    | A    | 900719925474099.2    | Amp  |
| Rated current (RMS) (0.0000000000000005% duty cycle)   | A    | 1801439850948198.4   | Amp  |
| Rated current (RMS) (0.00000000000000025% duty cycle)  | A    | 3602879701896396.8   | Amp  |
| Rated current (RMS) (0.0000000000000001% duty cycle)   | A    | 7205759403792793.6   | Amp  |
| Rated current (RMS) (0.00000000000000005% duty cycle)  | A    | 14411518807585587.2  | Amp  |
| Rated current (RMS) (0.000000000000000025% duty cycle) | A    | 28823037615171174.4  | Amp  |
| Rated current (RMS) (0.00000000000000001% duty cycle)  | A    | 57646075230342348.8  | Amp  |
| Rated current (RMS) (0.000000000000000005% duty cycle) | A    | 115292150460684697.6 | Amp  |

# Specification Sheet

## Electrical Characteristics

ELECTRICAL CHARACTERISTICS (T<sub>A</sub> = 25°C unless otherwise noted)

| Characteristic                                                                                      | Symbol           | Min  | Max  | Unit |
|-----------------------------------------------------------------------------------------------------|------------------|------|------|------|
| <b>DC ELECTRICAL CHARACTERISTICS</b>                                                                |                  |      |      |      |
| Supply Current (Maximum, V <sub>DD</sub> = 5V, I <sub>DD</sub> = 0, T <sub>A</sub> = 25°C)          | I <sub>DD</sub>  | 33   | 100  | μA   |
| Supply Current (Maximum, V <sub>DD</sub> = 5V, I <sub>DD</sub> = 0, T <sub>A</sub> = 0°C to 70°C)   | I <sub>DD</sub>  | —    | 100  | μA   |
| Supply Current (Maximum, V <sub>DD</sub> = 5V, I <sub>DD</sub> = 0, T <sub>A</sub> = -40°C to 85°C) | I <sub>DD</sub>  | —    | 100  | μA   |
| Supply Current (Typical, V <sub>DD</sub> = 5V, I <sub>DD</sub> = 0, T <sub>A</sub> = 25°C)          | I <sub>DD</sub>  | 4.13 | —    | μA   |
| <b>AC ELECTRICAL CHARACTERISTICS</b>                                                                |                  |      |      |      |
| Setup Time (Minimum, t <sub>su</sub> )                                                              | t <sub>su</sub>  | 1.0  | 5    | ns   |
| Hold Time (Minimum, t <sub>h</sub> )                                                                | t <sub>h</sub>   | 0.5  | —    | ns   |
| Propagation Delay (Maximum, t <sub>pd</sub> )                                                       | t <sub>pd</sub>  | 7.0  | —    | ns   |
| <b>ANALOG ELECTRICAL CHARACTERISTICS</b>                                                            |                  |      |      |      |
| Offset Voltage (Maximum, V <sub>OS</sub> )                                                          | V <sub>OS</sub>  | —    | ±0.5 | mV   |
| Common-Mode Rejection Ratio (Minimum, CMRR)                                                         | CMRR             | 80   | —    | dB   |
| Supply Voltage Range (Minimum, V <sub>DD</sub> )                                                    | V <sub>DD</sub>  | 1.8  | —    | V    |
| Supply Voltage Range (Maximum, V <sub>DD</sub> )                                                    | V <sub>DD</sub>  | —    | 5.5  | V    |
| Supply Current (Maximum, I <sub>DD</sub> )                                                          | I <sub>DD</sub>  | —    | 100  | μA   |
| Supply Current (Typical, I <sub>DD</sub> )                                                          | I <sub>DD</sub>  | —    | 4.13 | μA   |
| Supply Current (Maximum, I <sub>DD</sub> )                                                          | I <sub>DD</sub>  | —    | 100  | μA   |
| Supply Current (Typical, I <sub>DD</sub> )                                                          | I <sub>DD</sub>  | —    | 4.13 | μA   |
| Supply Current (Maximum, I <sub>DD</sub> )                                                          | I <sub>DD</sub>  | —    | 100  | μA   |
| Supply Current (Typical, I <sub>DD</sub> )                                                          | I <sub>DD</sub>  | —    | 4.13 | μA   |
| <b>TEMPERATURE RANGE</b>                                                                            |                  |      |      |      |
| Operating Temperature Range (T <sub>A</sub> )                                                       | T <sub>A</sub>   | -40  | 85   | °C   |
| Storage Temperature Range (T <sub>STG</sub> )                                                       | T <sub>STG</sub> | -55  | 150  | °C   |
| Maximum Junction Temperature (T <sub>J</sub> )                                                      | T <sub>J</sub>   | —    | 150  | °C   |
| Maximum Power Dissipation (P <sub>D</sub> )                                                         | P <sub>D</sub>   | —    | 100  | mW   |
| Maximum Power Dissipation (P <sub>D</sub> )                                                         | P <sub>D</sub>   | —    | 100  | mW   |

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# Handling MOSFETs

MOSFETs are very sensitive to static electricity. Because of the very thin  $\text{SiO}_2$  layer between the external terminals and the layers of the device, any small electrical discharge can create an unwanted conduction.

## Protection

- Always transport in a static sensitive bag
- Always wear a static strap when handling MOSFETs
- Apply voltage limiting devices between the gate and source, such as back-to-back Zeners to limit any transient voltage.

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[illegible]

# CMOS Devices

CMOS (complementary MOSFET) uses a *p*-channel MOSFET and *n*-channel MOSFET; often on the same substrate as shown here.

The diagram illustrates a cross-section of a CMOS device. It features two MOSFETs on a common substrate. The left MOSFET is an n-channel MOSFET with a p-type substrate, a p-well, and an n+ channel. The right MOSFET is a p-channel MOSFET with an n-type substrate, an n-well, and a p+ channel. Both devices have gates, source, and drain regions. The gates are connected to a common gate line, and the sources are connected to a common source line. The drains are connected to a common drain line. The diagram also shows the electrical connections for the gates, sources, and drains, and the overall structure of the device.

**Advantages**

- Useful in logic circuit designs
- Higher input impedance
- Faster switching speeds
- Lower operating power levels

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## Summary Table

|                                                                                                                                                                                                                                        |                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $I_{D1} = I_{D2} = I_{D3} = I_{D4}$<br><br>$I_{D1} = I_{D2} = I_{D3} = I_{D4} = \left( \frac{W}{L} \right)_{PMOS} \frac{(V_{DD}/2 - V_{th})^2}{2}$ |  |
| $I_{D1} = I_{D2} = I_{D3} = I_{D4}$<br><br>$I_{D1} = I_{D2} = I_{D3} = I_{D4} = \left( \frac{W}{L} \right)_{NMOS} \frac{(V_{DD}/2 - V_{th})^2}{2}$ |  |
| $I_{D1} = I_{D2} = I_{D3} = I_{D4}$<br><br>$I_{D1} = I_{D2} = I_{D3} = I_{D4} = \left( \frac{W}{L} \right)_{PMOS} \frac{(V_{DD}/2 - V_{th})^2}{2}$ |  |

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